

Distribution of Y_i in SLR

$$\begin{aligned}\mathbb{E}[Y_i] &= \mathbb{E}[\beta_0 + \beta_1 x_i + \epsilon_i] = \beta_0 + \beta_1 x_i + \mathbb{E}[\epsilon_i] \quad \text{Note: } \epsilon \sim N(0, \sigma^2) \\ &= \beta_0 + \beta_1 x_i\end{aligned}$$

$$\text{Var}(Y_i) = \text{Var}(\beta_0 + \beta_1 x_i + \epsilon_i) = \text{Var}(\epsilon_i) = \sigma^2$$

Since ϵ_i has a normal distribution, so does Y_i . Thus, the distribution of Y_i is

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Note that the Y_i 's are independent but NOT identically distributed since the means are changing through x_i

MLEs of β_0 and β_1 in SLR

$$\begin{aligned}f(y_i|\beta) &= (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{1}{2}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)^2\right\} \\ L(\beta) &= \prod_{i=1}^n f(y_i|\beta) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\sum_{i=1}^n \frac{1}{2}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)^2\right\} \\ \mathcal{L}(\beta) &= -\frac{n}{2} \log 2\pi\sigma^2 - \sum_{i=1}^n \frac{1}{2}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)^2\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\beta)}{\partial \beta_0} &= 0 - \left[\frac{1}{2\sigma^2} \left(2 \sum (y_i - \beta_0 - \beta_1 x_i) \right) (-1) \right] \\ &= \frac{1}{\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i) \\ \implies 0 &= \sum y_i - n\beta_0 - \beta_1 \sum x_i \\ \implies n\beta_0 &= n\bar{y} - \beta_1 \sum x_i \\ \implies \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\beta)}{\partial \beta_1} &= 0 - \left[\frac{1}{2\sigma^2} \left(2 \sum (y_i - \beta_0 - \beta_1 x_i) \right) (-x_i) \right] \\ &= \frac{1}{\sigma^2} \sum (y_i x_i - \beta_0 x_i - \beta_1 x_i^2) \\ \implies 0 &= \sum y_i x_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 \\ \implies 0 &= \sum y_i x_i - (\bar{y} - \beta_1 \bar{x}) \sum x_i - \beta_1 \sum x_i^2 \\ \implies 0 &= \sum y_i x_i - \bar{y} \sum x_i + \beta_1 \bar{x} \sum x_i - \beta_1 \sum x_i^2 \\ \implies \bar{y} n \bar{x} - \sum y_i x_i &= \beta_1 \left(n \bar{x}^2 - \sum x_i^2 \right) \quad \text{Multiply by } (-1) \\ \implies \sum y_i x_i - n \bar{x} \bar{y} &= \beta_1 \left(\sum x_i^2 - n \bar{x}^2 \right) \\ \implies \hat{\beta}_1 &= \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{XY}}{S_{XX}}\end{aligned}$$

Estimates of β_0 and β_1 by minimizing Q

$$Q = \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right)^2$$

$$\begin{aligned} \frac{\partial Q}{\partial \beta_0} &= 2 \sum (y_i - \beta_0 - \beta_1 x_i) (-1) = -2 \sum (y_i - \beta_0 - \beta_1 x_i) \\ &\implies 0 = \sum y_i - n\beta_0 - \beta_1 \sum x_i \\ &\text{Same as MLE so} \\ &\implies \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{aligned}$$

$$\begin{aligned} \frac{\partial Q}{\partial \beta_1} &= 2 \sum (y_i - \beta_0 - \beta_1 x_i) (-x_i) = -2 \sum (y_i - \beta_0 - \beta_1 x_i) (x_i) \\ &\implies 0 = \sum x_i y_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 \\ &\implies 0 = \sum y_i x_i - (\bar{y} - \beta_1 \bar{x}) \sum x_i - \beta_1 \sum x_i^2 \\ &\text{Same as MLE so} \\ &\implies \hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} \end{aligned}$$

Sampling Distribution of $\hat{\beta}_1$

Recall,

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x}) y_i - \sum (x_i - \bar{x}) \bar{y}}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x}) y_i}{S_{XX}} = \sum c_i y_i \quad \text{where } c_i = \frac{x_i - \bar{x}}{S_{XX}}$$

Note that the y_i 's are normally distributed. Thus $\hat{\beta}_1$ is a linear combination of normal random variables and thus is also normal.

$$\begin{aligned} \mathbb{E}[\hat{\beta}_1] &= \mathbb{E}\left[\sum c_i y_i\right] = \sum c_i \mathbb{E}[y_i] \\ &= \sum c_i (\beta_0 + \beta_1 x_i) \\ &= \beta_0 \sum \frac{x_i - \bar{x}}{S_{XX}} + \beta_1 \sum \frac{(x_i - \bar{x}) x_i}{S_{XX}} \quad \text{Note that } \sum (x_i - \bar{x}) = 0 \\ &= \beta_1 \sum \frac{(x_i - \bar{x}) x_i}{S_{XX}} \\ &= \frac{\beta_1}{S_{XX}} \sum (x_i - \bar{x}) x_i \end{aligned}$$

Lets work with just $\sum (x_i - \bar{x}) x_i$